Technical Brief: Targeting Agricultural Lime for Acid Croplands in MALAWI

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Outlook

1. Introduction
2. Spatial Distribution of Soil Acidity
3. Additional Production
4. Yield Loss
5. Yield Response
6. Lime Requirement
7. Economic Benefits
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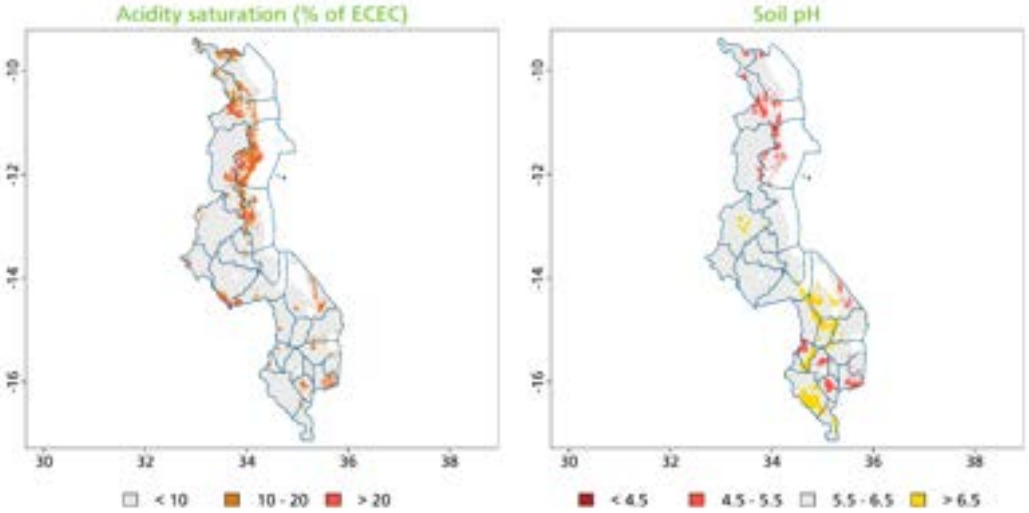
Introduction

- The slide deck provides a brief summary of **data-driven**, **ex-ante analysis** of soil acidity across **Malawi**, evaluating the scale of affected areas, potential crop yield impacts, and the expected benefits of lime application for proactive acid soil remediation.

- **Highlights**:
  - **Spatial Distribution of Acidic Cropland**: Identifies regions with high acidity levels impacting crop yields.
  - **Yield Loss**: Quantifies productivity losses in key crops due to acidic soils.
  - **Lime Requirements**: Estimates the lime amounts needed to remediate acid soils in these areas.
  - **Economic Potential**: Highlights estimated increases in production, production value and economic benefit from lime application.

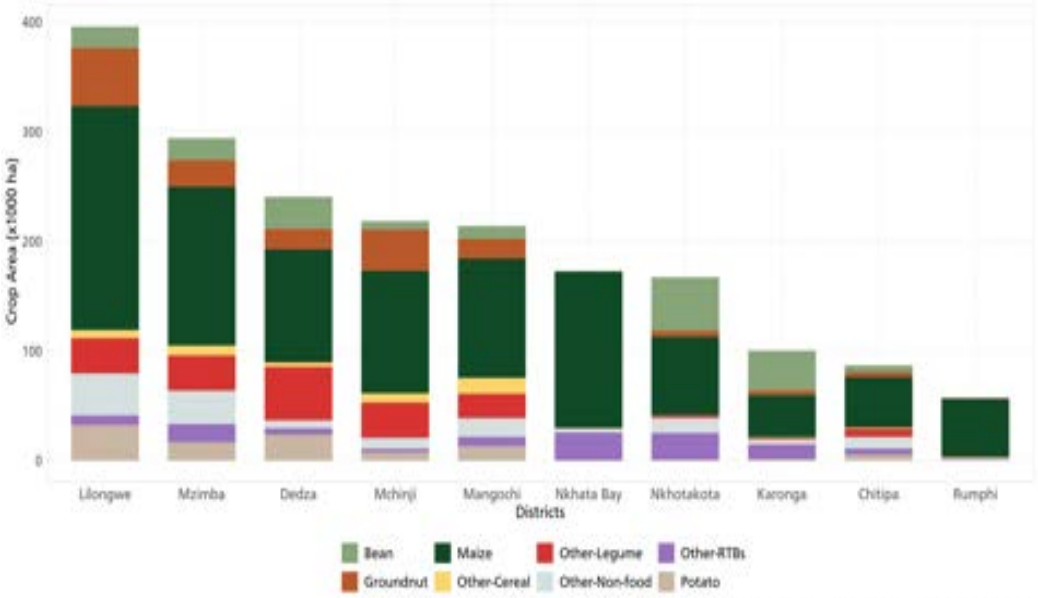
- Our goal is to provide policymakers, farmers, and stakeholders with actionable insights to prioritize areas with high potential for productivity gains and economic return from acid soil remediation.

Distribution of exchangeable acidity, as a % of the effective cation exchange capacity (ECEC), across Malawi



Top 10 Districts in Malawi with the highest acidic cropland

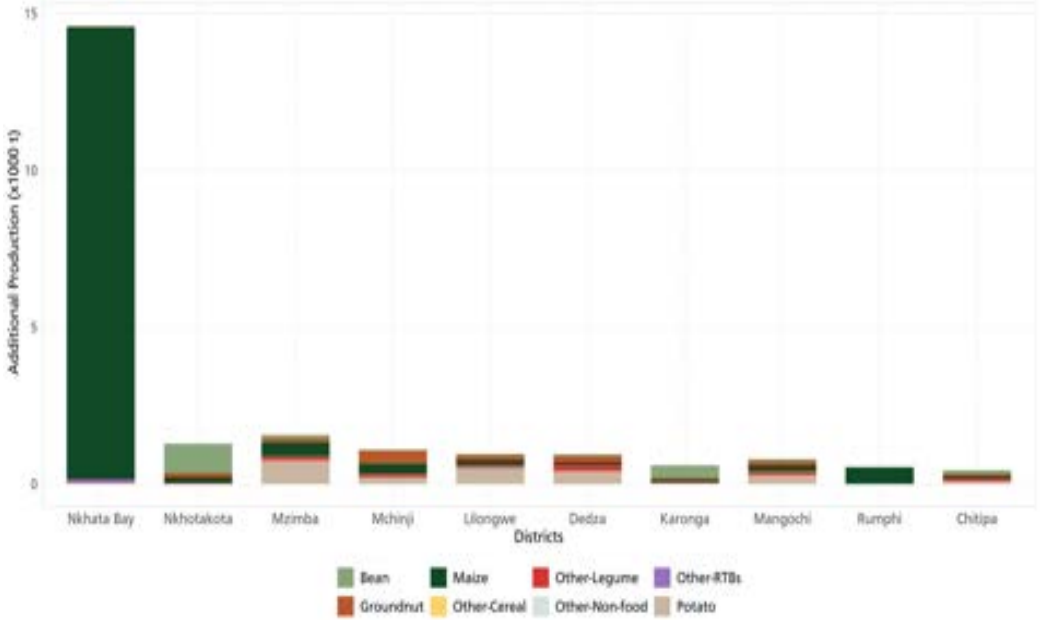
Total crop area in the top-10 most acidic Districts (not segregated by acidic cropland)



Total crop area in the top-10 most acidic Districts (not segregated by acidic cropland)

Districts	Bean	Groundnut	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Potato
Chitipa	6.05	4.73	46.09	1.22	7.25	10.64	5.47	5.50
Dedza	29.40	18.52	103.19	4.01	48.81	7.30	5.82	23.63
Karonga	36.59	4.24	38.04	0.80	1.45	4.36	13.71	1.16
Lilongwe	19.70	53.00	204.13	7.30	32.26	38.19	8.79	32.53
Mangochi	11.79	17.86	108.91	14.38	22.24	17.18	8.39	13.28
Mchinji	7.77	37.49	111.69	8.45	32.12	10.15	2.91	7.77
Mzimba	20.02	24.22	144.90	8.98	32.08	30.53	16.45	16.90
Nkhata Bay	0.48	0.17	142.57	0.12	0.28	3.75	25.48	0.17
Nkhotakota	49.01	5.37	71.21	0.25	2.43	13.76	24.52	0.69
Rumphi	0.62	0.67	52.61	0.31	0.76	1.57	0.55	0.34

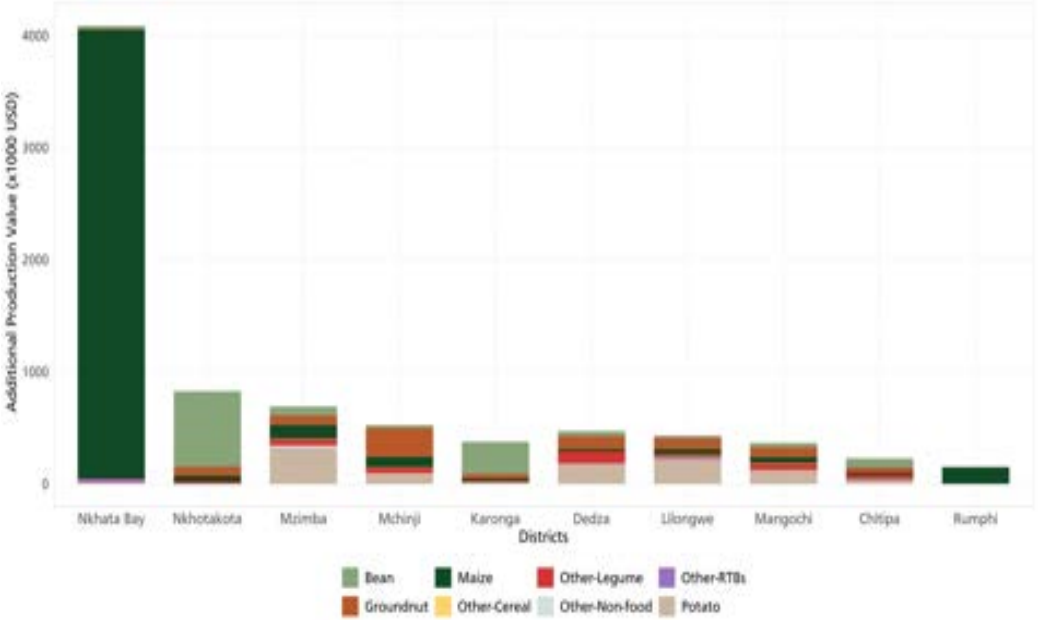
Potential for additional production (1000 tonnes) in top 10 acidic Districts in Malawi



Potential for additional production (1000 tonnes) in top 10 acidic Districts in Malawi

Districts	Bean	Groundnut	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Potato
Chitipa	0.11	0.07	0.07	-	0.08	-	0.01	0.09
Dedza	0.06	0.19	0.08	-	0.20	-	-	0.42
Karonga	0.40	0.06	0.07	-	0.03	-	-	0.03
Lilongwe	0.03	0.16	0.16	0.01	0.06	-	0.03	0.50
Mangochi	0.06	0.14	0.18	0.02	0.11	-	-	0.29
Mchinji	0.03	0.42	0.32	0.01	0.09	-	-	0.21
Mzimba	0.11	0.14	0.43	0.01	0.12	0.01	0.01	0.74
Nkhata Bay	0.04	0.01	14.40	-	0.01	-	0.11	0.03
Nkhotakota	0.93	0.13	0.18	-	0.04	-	-	-
Rumphi	-	-	0.53	-	-	-	-	-

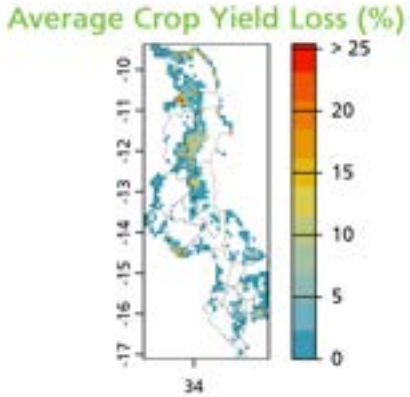
Potential for additional production value (1000 US$) in top 10 acidic Districts in Malawi

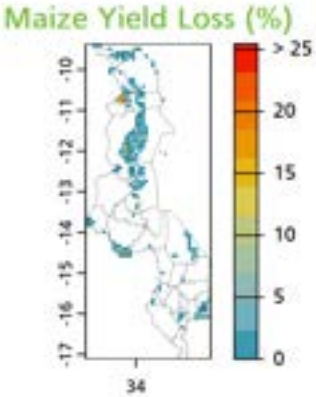


Potential for additional production value (1000 US$) in top 10 acidic Districts in Malawi

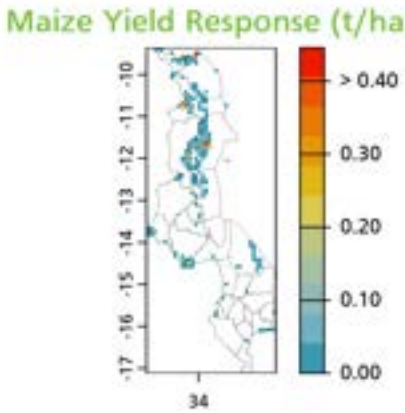
Districts	Bean	Groundnut	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Potato
Chitipa	81.8	45.7	20.1	0.6	40.1	-	3.6	37.6
Dedza	46.0	116.0	21.5	1.1	104.0	3.4	1.0	178.9
Karonga	289.1	39.0	20.4	0.3	13.4	-	0.1	14.5
Lilongwe	23.2	99.3	44.7	5.1	28.7	8.7	8.0	214.6
Mangochi	40.4	88.4	50.7	5.6	56.5	0.6	0.1	124.5
Mchinji	24.7	260.0	89.4	4.2	50.6	5.4	0.1	90.8
Mzimba	80.2	85.4	119.8	4.8	57.7	22.1	1.9	319.0
Nkhata Bay	27.6	9.2	4001.6	0.3	6.8	-	29.7	11.0
Nkhotakota	672.9	82.5	50.2	-	21.8	-	-	0.7
Rumphi	2.5	1.0	147.1	0.1	0.7	0.1	-	1.2

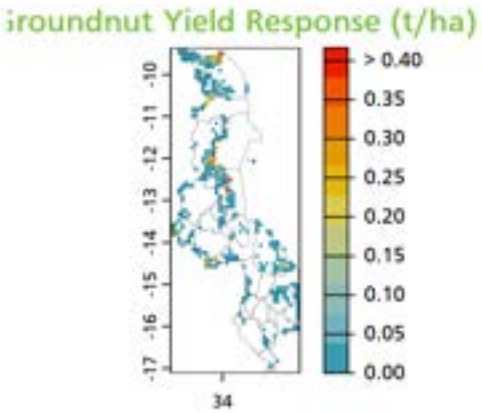
Estimated yield loss due to soil acidity in Malawi



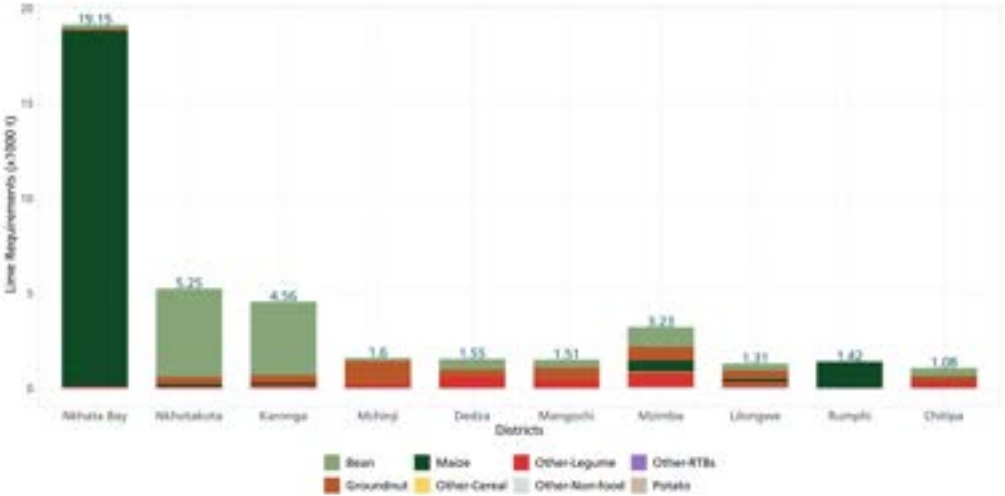


Yield response to lime application in Malawi





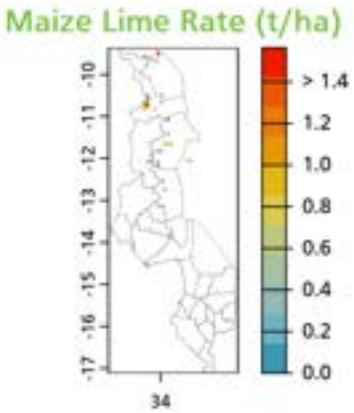
Lime requirements (1000 tonnes) for acid soil remediation top 10 acidic Districts in Malawi

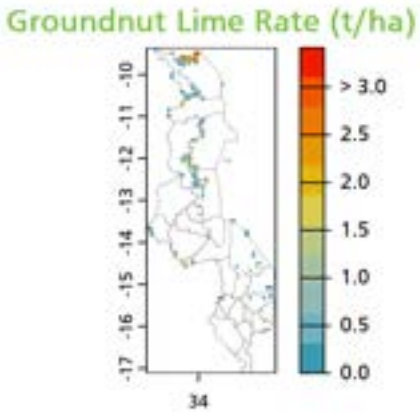


Lime requirements (1000 tonnes) for acid soil remediation in top 10 acidic Districts in Malawi

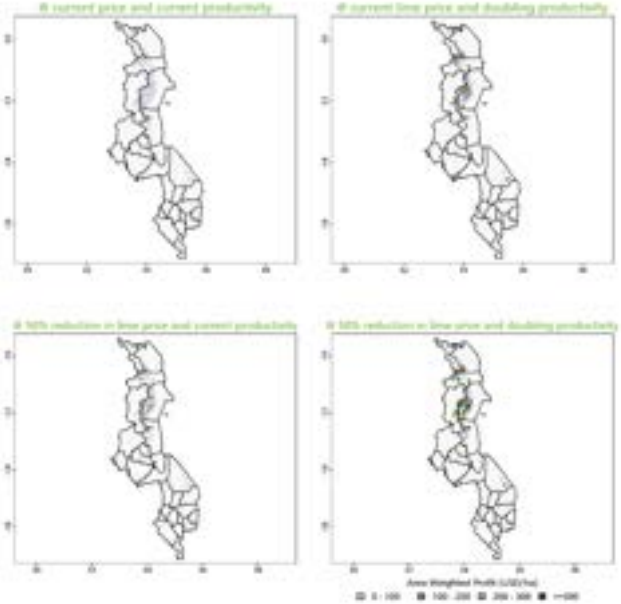
Districts	Bean	Groundnut	Maize	Other-Cereal	Other-Legume	Other-RTBs	Potato	Other-Non-food
Chitipa	0.41	0.28	0.04	-	0.33	-	0.01	-
Dedza	0.56	0.37	-	-	0.61	-	-	-
Karonga	3.86	0.40	0.12	0.01	0.16	-	0.01	-
Lilongwe	0.37	0.46	0.20	0.05	0.18	0.01	0.04	-
Mangochi	0.42	0.65	-	-	0.44	-	-	-
Mchinji	0.18	1.22	-	-	0.20	-	-	-
Mzimba	1.03	0.73	0.59	0.05	0.74	-	0.08	-
Nkhata Bay	0.22	0.10	18.73	-	0.09	-	-	-
Nkhotakota	4.66	0.39	0.13	-	0.08	-	-	-
Rumphi	0.01	0.01	1.39	0.01	-	-	-	0

Lime rate for Maize and Groundnut in Malawi



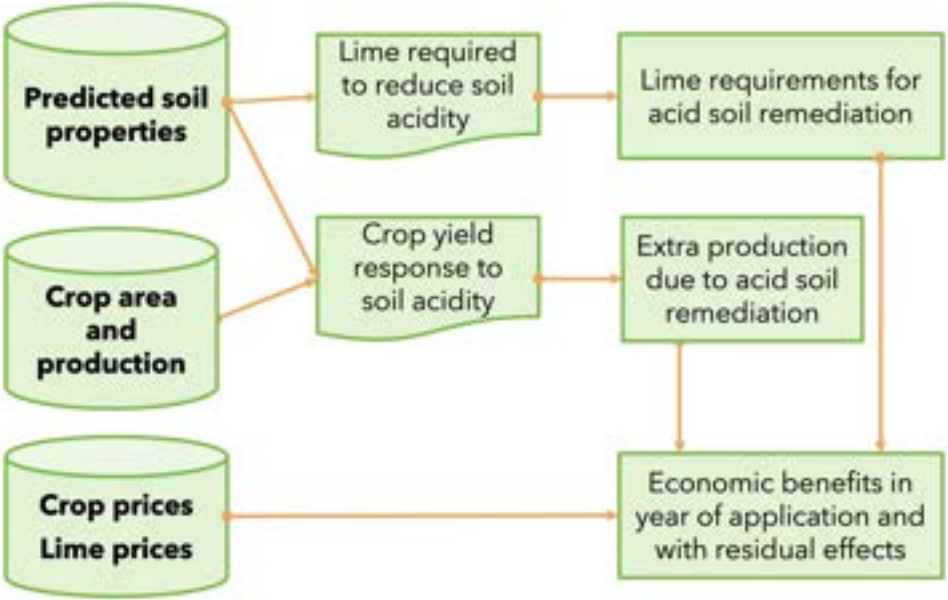


Spatial Distribution of crop-area weighted average profitability (US$/ha) Under Varying Prices and Productivity assumptions in Malawi.



Appendix

Modeling approach



Modeling approach: Three-Step Approach

- **Lime Requirements Estimation**:
  - Estimated using Aramburu Merlos et al. (2023) lime requirement model.
  - Calculated lime rates to achieve crop-specific Target Acidity Saturation (TAS).
  - Applied to gridded soil data on exchangeable acidity and ECEC from Hengl et al. (2017).
  - Converted lime rates to tons of $CaCO_3/ha$, assuming a 20 cm soil depth and using soil bulk density data.
- **Crop Yield Response Estimation**:
  - Plateau-linear decay models representing yield loss due to acidity (estimated current yield losses (% of maximum yield) due to high acidity saturation).
- **Economic Benefit Estimation**:
  - Return is calculated as the additional yield from liming multiplied by the median crop price.
  - Cost is determined by multiplying the estimated lime rate by a fixed lime price.

For further details on the methodological approach, visit www.acidsoils.africa.

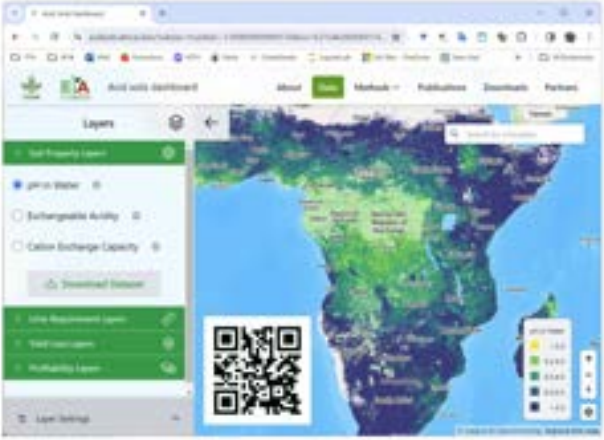
Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after acidic soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

For details, visit www.acidsoils.africa.

Public goods for investment guidance

- **Acid soils dashboard** for spatial targeting & priority setting
  - Interactive; downloadable data
- **Profitability explorer**
  - What ROI to liming vary by input/output price assumptions?
- **Acidification Explorer**
  - Coming soon
- **Pooled, curated FAIR data**
  - Crop responses to liming
- **Analytical workflows**
  - Reproducible, adaptable scripts
  - R package for lime requirements



https://www.acidsoils.africa/

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  - Ministry of Agriculture, Ethiopia
  - Zambian Agricultural Research Institute (ZARI)

Data and code availability:

- The data used in this analysis are available from: www.acidsoils.africa.
- R scripts developed for data manipulation, analysis and visualization are available through the GitHub repository: https://github.com/EiA2030-ex-ante/gaia-exante-framework.

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